

Syna Malhan

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SUMMARY

Computer Science student (4.0 GPA) with hands-on experience in machine learning, computer vision, and AI system development. Strong background in deep learning, OCR, RAG systems, and data-driven applications using Python, Transformers, YOLO, and Neo4j. Proven ability to build end-to-end AI pipelines, fine-tune models, and deliver real-world impact through human-centered design and technical rigor.

EDUCATION

Arizona State University, Tempe, AZ — Bachelor of Science in Computer Science | Minor in Data Science

GPA: 4.0 | Graduation: 2027

Awards: Dean's List (4x), New American University Scholar, Grace Hopper Celebration Scholar

SKILLS

Machine Learning & AI: Transformers (BERT, RoBERTa, Donut), YOLO, U-Net, XGBoost, LSTM, ARIMA, Prophet, RAG, LangGraph, FastSAM, SAM

Data Science & Analytics: Pandas, NumPy, scikit-learn, SQL, Matplotlib, Streamlit, Plotly, FRED API

Computer Vision: OpenCV, TensorFlow, Keras, Ultralytics, FastSAM, OCR (Donut, AWS OCR, Azure OCR)

Tools & Frameworks: Python, React, FastAPI, Flask, Neo4j, Hugging Face, Ollama, Firebase, Docker

Platforms & Deployment: Streamlit, Vercel, Supabase, GitHub Pages, Apple Developer Tools

Collaboration: Git, Agile, Scrum, Figma, Postman, VSCode, Zoom

EXPERIENCE

Data Science Intern | *Infinite Uptime* | Jun 2025 - Present

- Analyzing industrial process data to identify operational inefficiencies, detect anomalies, and generate actionable insights for predictive maintenance.
- Leveraging tools like Python, Pandas, NumPy, and scikit-learn to clean, preprocess, and model real-world time-series sensor data from manufacturing systems.
- Creating dashboards and reports using Power BI or Matplotlib to visualize trends, downtimes, and KPIs, helping cross-functional teams make data-driven decisions.

GenAI Intern | *Enrest and Young* | Jun 2025 - Aug 2025

- Collaborated on building visualization-based AI systems for enterprise applications using Retrieval-Augmented Generation (RAG).
- Utilized Neo4j for graph-based data modeling and relationship mapping to improve spatial and contextual insights.
- Developed interactive dashboards and visual workflows to demonstrate AI outputs and knowledge graph structures.
- Participated in virtual sprint planning and team reviews, contributing to project documentation and technical presentations.

Cloud Front End Developer | *ASU AI Cloud Innovation Center* | Aug 2024 - Present

- Collaborated with cross-functional teams to design UI for public sector applications.
- Created wireframes and data visualizations to illustrate spatial usage and cloud workflows.
- Regularly reported progress via virtual meetings and managed tasks using cloud-based tools.

ML and Data Analytics Intern | *Ripik.AI* | Jul 2024 - Aug 2024

- Trained object detection models (YOLO), improving accuracy for industrial use cases.
- Developed OCR models and boosted form recognition performance by 25%.
- Produced dashboards and visual analytics for client reports.

Intern in Digitalization | *Jindal Steel and Power Ltd* | May 2024 - Jul 2024

- Developed internal data dashboards with AngularJS and Spring Boot.
- Enhanced visual clarity of operational layouts by redesigning space-related UI elements.
- Integrated performance metrics and reports into stakeholder presentations.

Software Engineering Fellow | *Headstarter AI* | Jul 2024 - Sep 2024

- Built data-rich, interactive AI apps using React, OpenAI APIs, and visualization libraries.
- Led peer reviews and presentation sessions via Zoom, enhancing team collaboration remotely.
- Led 4+ engineering fellows in full-stack development, with coaching from Amazon, Bloomberg, and Capital One engineers.

PROJECTS

Wildlife Monitoring and Conservation — YOLO, Faster R-CNN, SAM, FastSAM, Python

- Automated species detection and segmentation from camera trap & satellite imagery.
- Used ensemble models (YOLO + FastSAM) to reach 92%+ detection accuracy in variable lighting.
- Enabled 24/7 monitoring and alerting of endangered species movement patterns.

Slab Sizing — YOLO, OpenCV, Python

- Automated detection of slab dimensions using object detection and image processing.
- Calibrated YOLO model to recognize contours and measure length/width accurately.

Text-Optimized Image Generation Model — HuggingFace, Diffusers, PyTorch, LoRA, Stable Diffusion

- Fine-tuned Stable Diffusion with LoRA adapters for high-clarity text generation in images.
- Achieved significant improvements in OCR accuracy on generated visual forms.
- Created custom dataset with labels, signs, and form layouts for supervised training.

Credit Risk RAG Model — LangGraph, Neo4j, Streamlit, Ollama, Pandas

- Built a retrieval-augmented reasoning system using Neo4j for credit policy graphs.
- Delivered explainable credit decisions through LangGraph and Ollama inference.
- Built a user-facing dashboard in Streamlit for interactive exploration of risk factors.

Plate Mill Finished Goods Forecasting — XGBoost, ARIMA, Prophet, Random Forest, Python

- Built ensemble and time series models to forecast weekly dispatch volumes.
- Improved forecasting accuracy by 30%, enhancing supply chain decisions.

Portfolio Analyzer & Stock Prediction Dashboard — Streamlit, LSTM, Random Forest, Plotly

- Built a real-time portfolio tracker with ML-based stock forecasting.
- Integrated LSTM for sequential data and RF for trend smoothing.